Gentzen-type Proof Systems for Non-classical Logics

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Background

Hilbert Calculi

A Hilbert-style system

- Axioms:
  1. $A \supset (B \supset A)$
  2. $(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$
  3. $(\neg A \supset \neg B) \supset (B \supset A)$
- Rule: From A and $A \supset B$ infer B

A Hilbert-style proof of $p \supset p$

1. $p \supset ((p \supset p) \supset p)$
2. $(p \supset ((p \supset p) \supset p)) \supset ((p \supset (p \supset p)) \supset (p \supset p))$
3. $(p \supset (p \supset p)) \supset (p \supset p)$
4. $p \supset (p \supset p)$
5. $p \supset p$

> *The formalization of logical deduction . . . is rather far removed from the forms of deduction used in practice in mathematical proofs.*

Gentzen 1934

Natural Deduction

> *I intended . . . to set up a formal system which came as close as possible to actual reasoning.*

> *The result was a "calculus of natural deduction." ("NJ" for intuitionist, "NK" for classical predicate logic.)*

Gentzen 1934

$$\dfrac{X \vee (Y \,\&\, Z) \qquad \dfrac{\dfrac{\overset{1}{X}}{X \vee Y}\vee\text{-}I \quad \dfrac{\overset{1}{X}}{X \vee Z}\vee\text{-}I}{(X \vee Y)\,\&\,(X \vee Z)}\,\&\text{-}I \qquad \dfrac{\dfrac{\dfrac{\overset{1}{Y \,\&\, Z}}{Y}\,\&\text{-}E}{X \vee Y}\vee\text{-}I \quad \dfrac{\dfrac{\overset{1}{Y \,\&\, Z}}{Z}\,\&\text{-}E}{X \vee Z}\vee\text{-}I}{(X \vee Y)\,\&\,(X \vee Z)}\,\&\text{-}I}{(X \vee Y)\,\&\,(X \vee Z)}\vee\text{-}E\,1}{(X \vee (Y \,\&\, Z)) \supset ((X \vee Y)\,\&\,(X \vee Z))}\supset\text{-}I\,2$$

The Subformula Property

> *No concepts enter into the proof other than those contained in its final result, and their use was therefore essential to the achievement of that result.*

> *For this the natural calculus proved unsuitable . . . it does so only with respect to its intuitionist form. . .*

Gentzen 1934

Importance of the subformula property:

- Decidability (of the propositional fragment)
- Consistency (e.g. $\bot$ is not derivable)

Sequent Calculus

> *I shall develop a new calculus of logical deduction containing all the desired properties in both their intuitionist and their classical form. ("LJ" for intuitionist, "LK" for classical predicate logic).*

Sequents

- Motivation: subformula property
- The calculi will manipulate formulas of the form $A_1 \wedge \ldots \wedge A_n \supset B_1 \vee \ldots \vee B_m$
- Replace $\supset$ with "$\Rightarrow$", $\wedge$ on the left and $\vee$ on the right with ",".
- Add structural rules for "," and "$\Rightarrow$"

Sequent Calculus

The Propositional Fragment of **LK** [Gentzen 1934]

$$(id)\ \frac{}{A \Rightarrow A} \qquad (cut)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \Delta} \qquad (weak)\ \frac{\Gamma \Rightarrow \Delta}{\Gamma, \Gamma' \Rightarrow \Delta, \Delta'}$$

$$(\neg \Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, \neg A \Rightarrow \Delta} \qquad (\Rightarrow \neg)\ \frac{\Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \neg A, \Delta}$$

$$(\wedge \Rightarrow)\ \frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad (\Rightarrow \wedge)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}$$

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The Cut-elimination Theorem ("Hauptsatz")

> *Every . . . LK-derivation can be transformed into another . . . LK-derivation with the same endsequent, in which no cuts occur.*

Sequent Calculus

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Analyticity ("The Subformula Property")

> *Corollary of the Hauptsatz (subformula property): In an . . . LK-derivation without cuts, all occurring . . . formulae are subformulae of . . . the endsequent.*

Sequent Calculus

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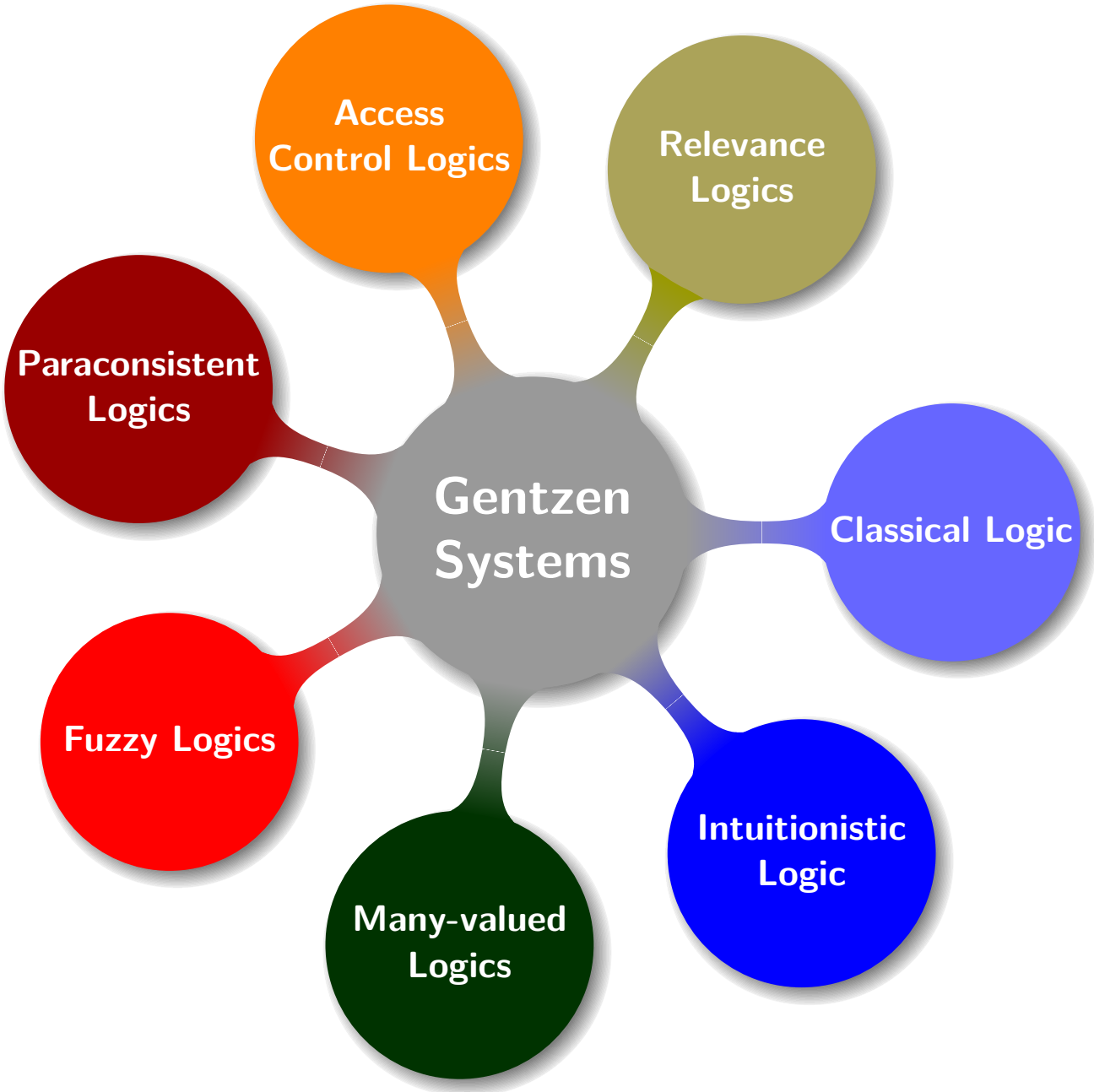
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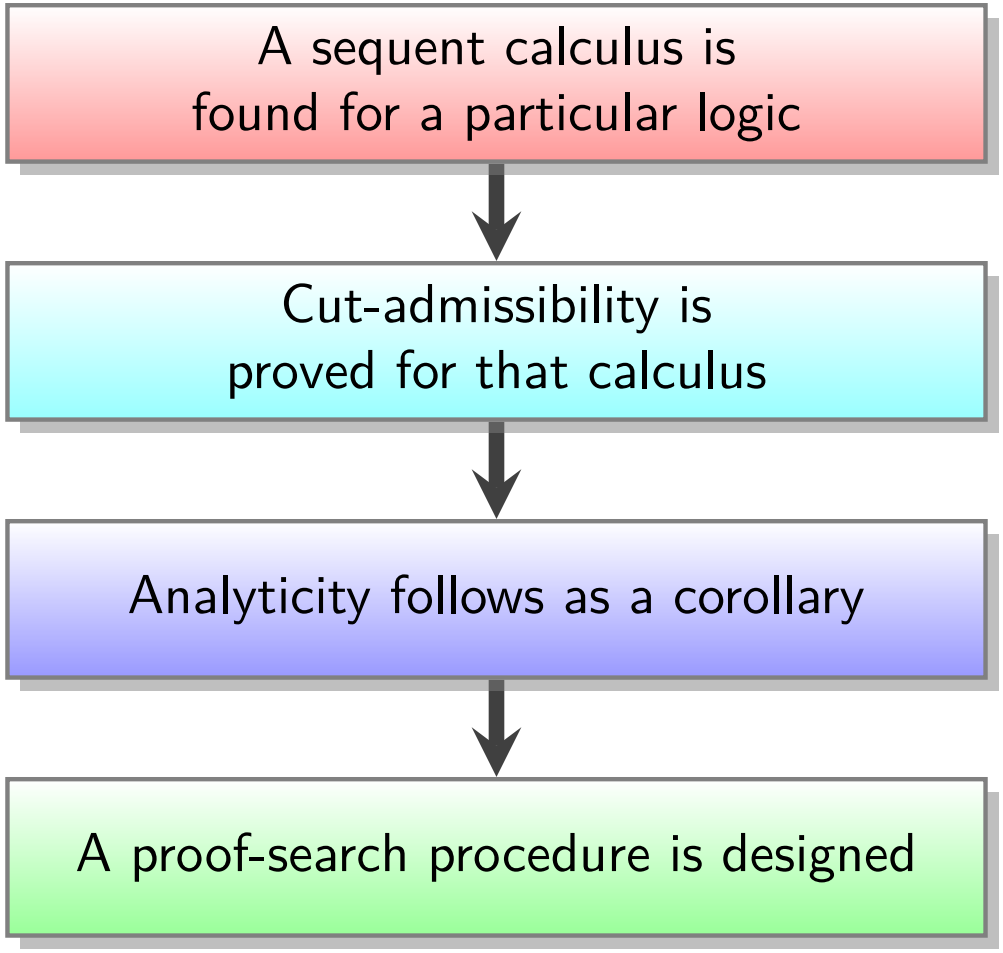
LK is ordered: premises are subformulas of **the** main formula

LK is ordered → (cut-admissibility → analyticity)

Beyond **LK**


Access Control Logics
Relevance Logics
Paraconsistent Logics
Gentzen Systems
Classical Logic
Fuzzy Logics
Intuitionistic Logic
Many-valued Logics

The Traditional Route



Problems

- Cut-admissibility proofs are complicated and error-prone
- Analyticity vs. cut-admissibility
- Efficient proof-search procedures require a great effort

More Analyticities

Calculus for FDE [Arieli, Avron 1998]

$$\frac{\Gamma, \neg A \Rightarrow \Delta \quad \Gamma, \neg B \Rightarrow \Delta}{\Gamma, \neg(A \wedge B) \Rightarrow \Delta} \qquad \frac{\Gamma \Rightarrow \neg A, \neg B, \Delta}{\Gamma \Rightarrow \neg(A \wedge B), \Delta}$$

- Not analytic, not ordered
- $\neg$-analytic, $\neg$-ordered
- Many calculi for non-classical logics only admit such weaker variants

$\circledcirc$-k-subformulas

A is an *immediate* $\circledcirc$-k-*subformula* of B if one of the followings hold:

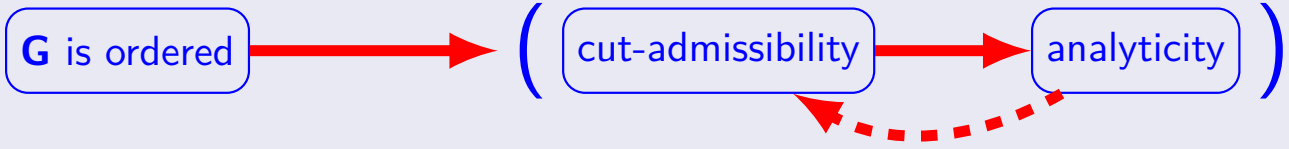
- $B = \diamond(B_1, \ldots, B_n)$ and $A = \bar{\circ} B_i$ for some n-ary connective $\diamond \notin \circledcirc$, $1 \leq i \leq n$, and $\circledcirc$-string $\bar{\circ}$ of length $\leq k$
- $A = \circ B$ for some $\circ \in \circledcirc$

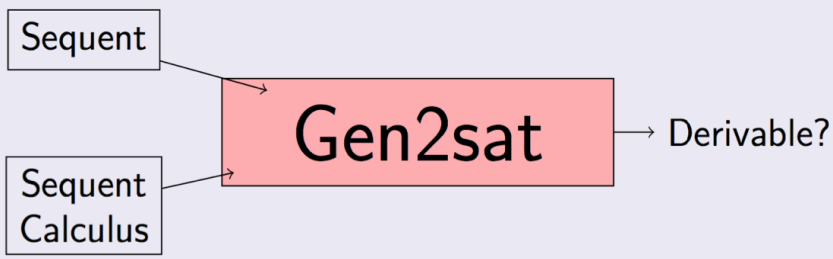
$\neg\neg p$ is a $\{\neg\}$-2-subformula of $p \wedge q$.

Results

Main Results





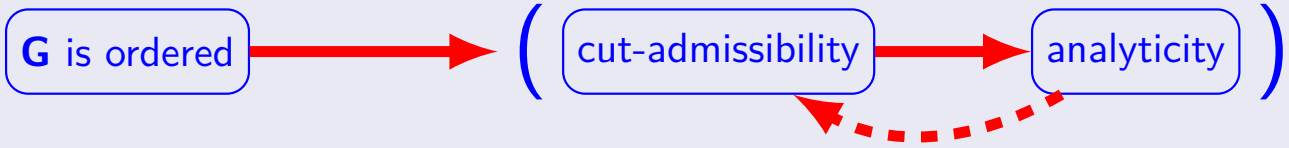


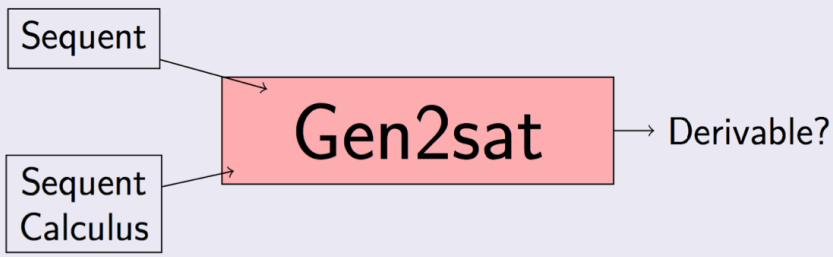
Families of Calculi

- Pure calculi
- Pure calculi + Modal operators
- Intuitionistic calculi

Main Results







Families of Calculi

- Pure calculi
- Pure calculi + Modal operators
- Intuitionistic calculi

Pure Calculi

- Propositional
- Sequents are pairs of finite sets
- Fully-structural (cut, weakening, id)
- No context-restriction on applications of rules

Examples

- Prototype example: **LK**
- Primal infon iogic
- Paraconsistent logics
- Three and four valued logics

$$\vdots$$

✓ $$\frac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \supset B, \Delta}$$

✗ $$\frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \supset B}$$

✗ $$\frac{\Gamma \Rightarrow A}{\Box\Gamma \Rightarrow \Box A}$$

Sufficient Criterion for Analyticity



Ordered and Directed Calculi

- **G** is ordered if in every rule, every formula from the premises is a proper subformula of some formula from the conclusion
- directed = ordered + one formula in the conclusion

Coherent Calculi (Generalizes Avron & Lev 2005)

G is coherent if for any two rules of the form

$$\frac{S_1}{\Rightarrow A} \qquad \frac{S_2}{B \Rightarrow}$$

and σ_1, σ_2 with $\sigma_1(A) = \sigma_2(B)$, we have $\sigma_1(S_1), \sigma_2(S_2) \vdash^{(cut)} \Rightarrow$

Theorem

Every coherent directed pure calculus is analytic.

Sufficient Criterion for Analyticity



Ordered and Directed Calculi

- **G** is ⊙-k-ordered if in every rule, every formula from the premises is a proper ⊙-k-subformula of some formula from the conclusion
- ⊙-k-directed = ⊙-k-ordered + one formula in the conclusion

Coherent Calculi (Generalizes Avron & Lev 2005)

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Theorem

Every coherent ⊙-k-directed pure calculus is ⊙-k-analytic.

Analyticity-by-construction



- Consider the following applications of $\dfrac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \supset B, \Delta}$:

$$\checkmark \ \frac{A \Rightarrow A}{\Rightarrow A \supset A} \qquad \checkmark \ \frac{A, A \wedge B \Rightarrow A, B}{A \Rightarrow (A \wedge B) \supset A, B} \qquad \times \ \frac{B \vee C, A \Rightarrow B}{B \vee C \Rightarrow A \supset B}$$

- These applications constitute new (weaker) rules:

$$\frac{\Gamma, A \Rightarrow A, \Delta}{\Gamma \Rightarrow A \supset A, \Delta} \qquad \frac{\Gamma, A, A \wedge B \Rightarrow A, B, \Delta}{\Gamma, A \Rightarrow (A \wedge B) \supset A, B, \Delta} \qquad \frac{\Gamma, B \vee C, A \Rightarrow B, \Delta}{\Gamma, B \vee C \Rightarrow A \supset B, \Delta}$$

Theorem

Let **G** be a coherent directed calculus. Every calculus whose rules are ordered applications of **G** is analytic.

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Theorem

Let **G** be a coherent $\circledcirc$-k-directed calculus. Every calculus whose rules are $\circledcirc$-k-ordered applications of **G** is analytic.

Analyticity-by-construction: Examples

The Atomic Paraconsistent Logic P_1 [Sette 1973]

$$(\neg\Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, \neg A \Rightarrow \Delta}$$

$$\frac{\Gamma \Rightarrow \neg A, \Delta}{\Gamma, \neg\neg A \Rightarrow \Delta} \qquad \frac{\Gamma \Rightarrow A \wedge B, \Delta}{\Gamma, \neg(A \wedge B) \Rightarrow \Delta}$$

$$\frac{\Gamma \Rightarrow A \vee B, \Delta}{\Gamma, \neg(A \vee B) \Rightarrow \Delta} \qquad \frac{\Gamma \Rightarrow A \supset B, \Delta}{\Gamma, \neg(A \supset B) \Rightarrow \Delta}$$

- Paraconsistency applies only in the atomic level.
- $\nvdash_{P_1} p, \neg p \Rightarrow B$.
- $\vdash_{P_1} A, \neg A \Rightarrow B$ whenever A is compound.

Analyticity-by-construction: Examples

Calculus for Primal Infon Logic [Gurevich,Neeman '09]

$$(\wedge\Rightarrow)\ \frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad (\Rightarrow\wedge)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}$$

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- An extremely efficient propositional logic.
- One of the main logical engines behind DKAL (Distributed Knowledge Authorization Language).
- Provides a balance between expressivity and efficiency.

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Analyticity-by-construction: Examples

Extended Primal Infon Logic

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$$\frac{}{\Gamma \Rightarrow A \supset A, \Delta} \qquad \frac{}{\Gamma \Rightarrow B \supset (A \supset B), \Delta} \qquad \frac{}{\Gamma \Rightarrow (A \wedge B) \supset A, \Delta} \qquad \frac{}{\Gamma \Rightarrow (A \wedge B) \supset B, \Delta}$$

$$\frac{}{\Gamma, A \vee A \Rightarrow A, \Delta} \qquad \frac{}{\Gamma, A \vee (A \wedge B) \Rightarrow A, \Delta} \qquad \frac{}{\Gamma, (A \wedge B) \vee A \Rightarrow A, \Delta}$$

$$\frac{}{\Gamma, \bot \Rightarrow \Delta} \qquad \frac{}{\Gamma \Rightarrow \bot \supset A, \Delta} \qquad \frac{}{\Gamma, \bot \vee A \Rightarrow A, \Delta} \qquad \frac{}{\Gamma, A \vee \bot \Rightarrow A, \Delta}$$

Modal Operators

We study the extension of arbitrary pure calculi, with the following rules:

Modal Rules

$$(\mathrm{K})\frac{\Gamma \Rightarrow A}{\Box\Gamma \Rightarrow \Box A} \qquad (4)\frac{\Box\Gamma_1, \Gamma_2 \Rightarrow A}{\Box\Gamma_1, \Box\Gamma_2 \Rightarrow \Box A} \qquad (45)\frac{\Box\Gamma_1, \Gamma_2 \Rightarrow A, \Box\Delta}{\Box\Gamma_1, \Box\Gamma_2 \Rightarrow \Box A, \Box\Delta}$$

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T Rule

$$(\mathrm{T})\frac{\Gamma, A \Rightarrow \Delta}{\Gamma, \Box A \Rightarrow \Delta} \qquad S4 = \mathrm{K} + 4 + \mathrm{T} \qquad S5 = \mathrm{K} + \mathrm{B4} + \mathrm{T}$$

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Analyticity Survives



For every pure calculus **G** and every modal combination X, $\mathbf{G}_X = \mathbf{G} \cup \{X\}$.

Main Theorem

Let **G** be a pure calculus. If **G** is analytic, then so is $\mathbf{G}_X$.

Analyticity Survives



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Let **G** be a pure calculus. If **G** is $\circledcirc$-k-analytic, then so is $\mathbf{G}_X$.

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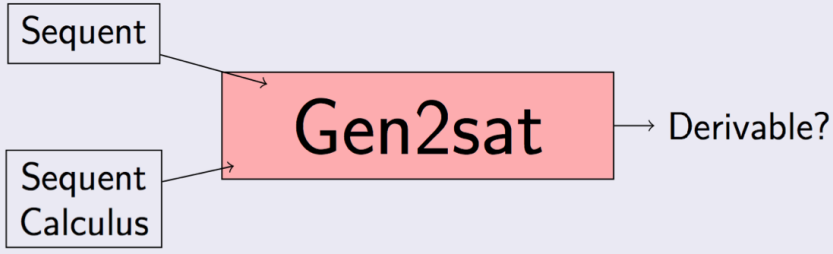
Examples

- Analyticity of calculi for classical modal logics follows from that of **LK**
- Quotations for primal infon logic: “q said"
- Adding modalities to three-valued logics or paraconsistent logics

Main Results







Families of Calculi

- Pure calculi
- Pure calculi + Modal operators
- Intuitionistic calculi

Cut-admissibility vs. Analyticity



Theorem

Every pure directed analytic calculus admits cut-admissibility.

Question

"But analyticity itself is usually proven using cut-admissibility!"

Well...

- Theoretical interest
- Simpler proofs of cut-admissibility
  - Employ existing criteria for analyticity
  - From analytic cut-admissibility to full cut-admissibility

Cut-admissibility vs. Analyticity



Theorem

Every pure $\circledcirc$-k-directed $\circledcirc$-k-analytic calculus admits cut-admissibility.

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- Simpler proofs of cut-admissibility
  - Employ existing criteria for analyticity
  - From analytic cut-admissibility to full cut-admissibility

Cut-admissibility vs. Analyticity



Theorem

1. Every pure directed analytic calculus admits cut-admissibility.
2. Every intuitionistic directed analytic calculus admits cut-admissibility.

The Intuitionistic Calculus $\mathbf{LJ}'$ [Maehara 1954, Takeuti 1975]

$$(\neg \Rightarrow) \quad \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, \neg A \Rightarrow \Delta} \qquad (\Rightarrow \neg) \quad \frac{\Gamma, A \Rightarrow}{\Gamma \Rightarrow \neg A}$$

$$(\wedge \Rightarrow) \quad \frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad (\Rightarrow \wedge) \quad \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}$$

$$(\vee \Rightarrow) \quad \frac{\Gamma, A \Rightarrow \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \vee B \Rightarrow \Delta} \qquad (\Rightarrow \vee) \quad \frac{\Gamma \Rightarrow A, B, \Delta}{\Gamma \Rightarrow A \vee B, \Delta}$$

$$(\supset \Rightarrow) \quad \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \supset B \Rightarrow \Delta} \qquad (\Rightarrow \supset) \quad \frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \supset B}$$

Cut-admissibility vs. Analyticity



Theorem

1. Every pure directed analytic calculus admits cut-admissibility.
2. Every intuitionistic directed analytic calculus admits cut-admissibility.

Intuitionistic Calculi

- Prototype example: **LJ**$'$
- (Paraconsistent) constructive logics (e.g., Nelson's logic N4)

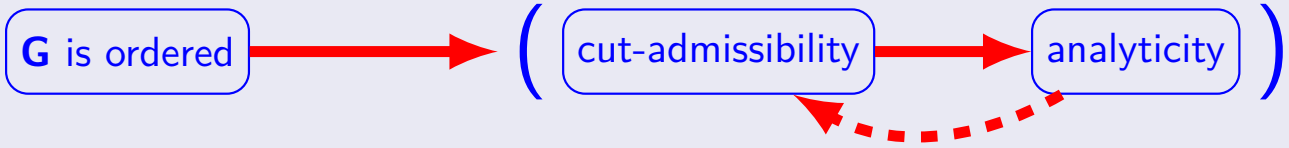
Beyond Pure and Intuitionistic

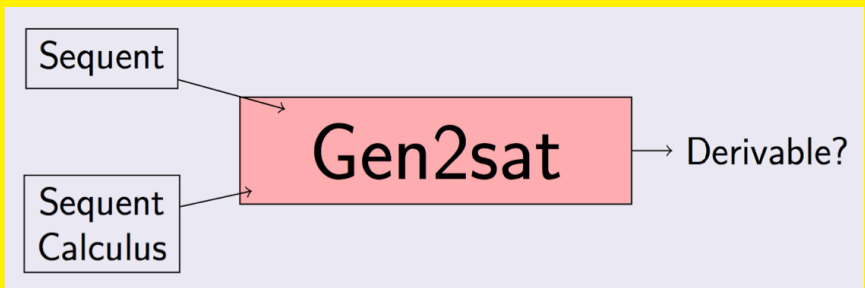
- Modal logics B and S5
- Calculi that include the following two rules:

$$(\Rightarrow \neg) \quad \frac{\Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \neg A, \Delta} \qquad (\Rightarrow \supset) \quad \frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \supset B}$$

Main Results



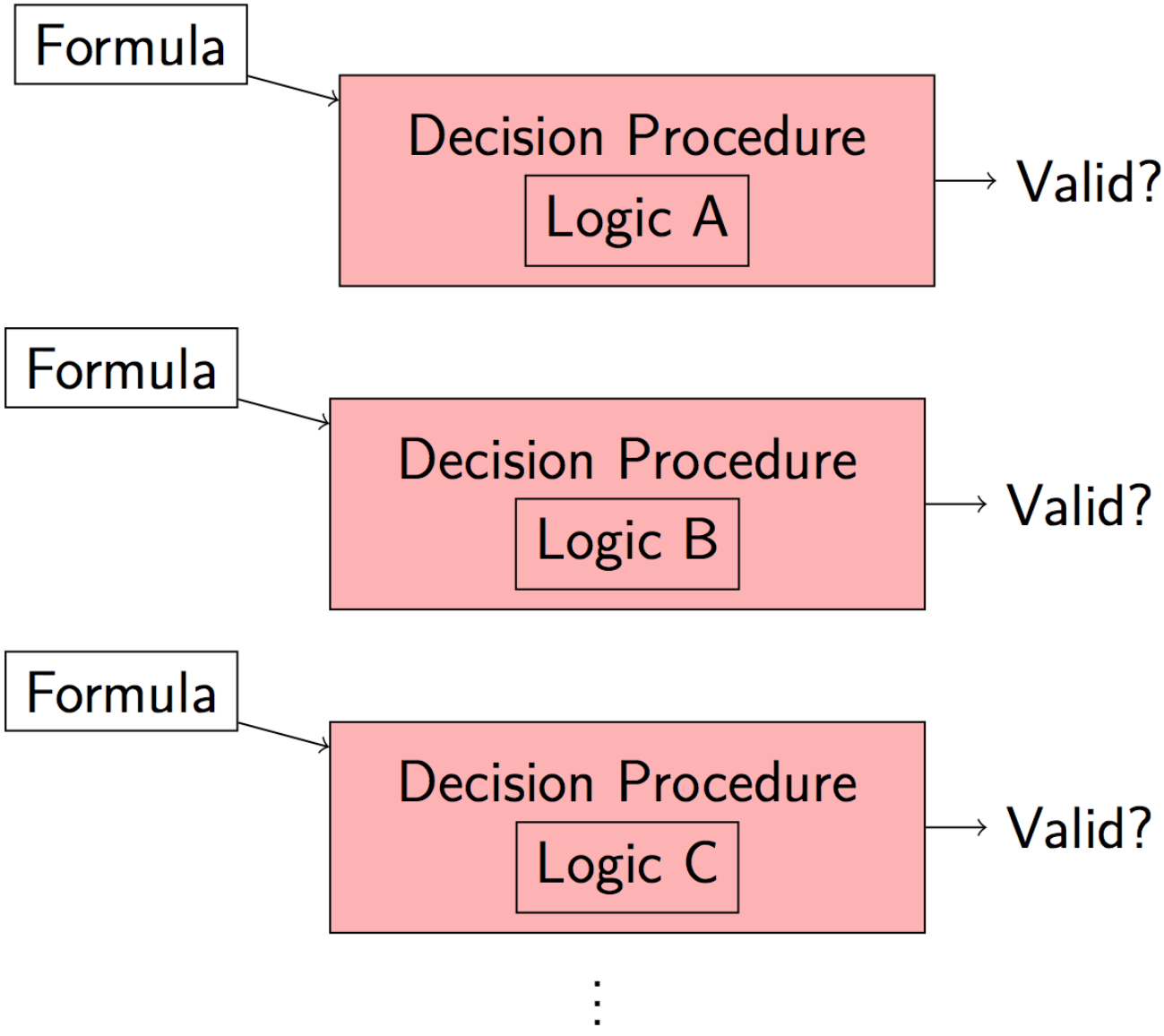




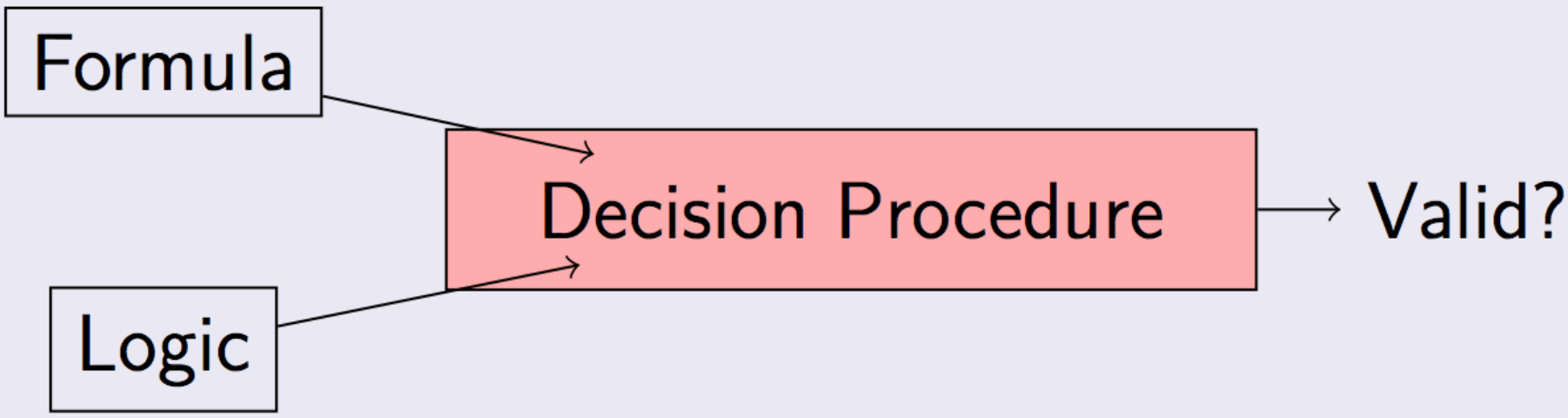
Families of Calculi

- Pure calculi
- Pure calculi + Modal operators
- Intuitionistic calculi

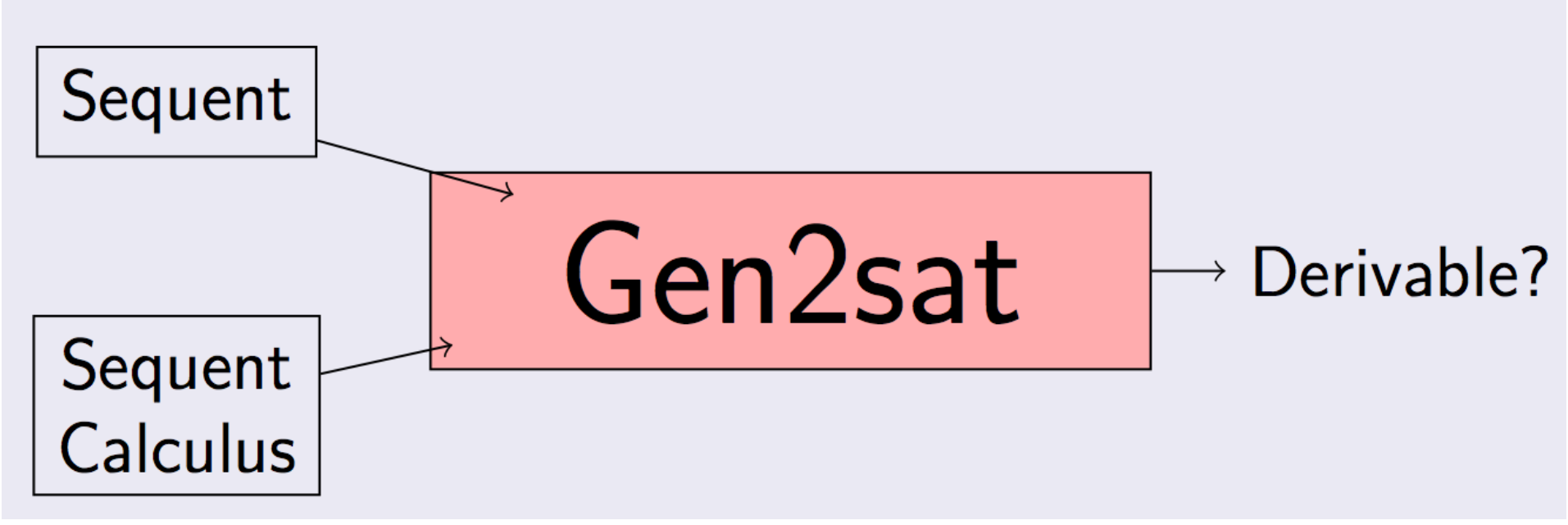
Uniform Decision Procedure



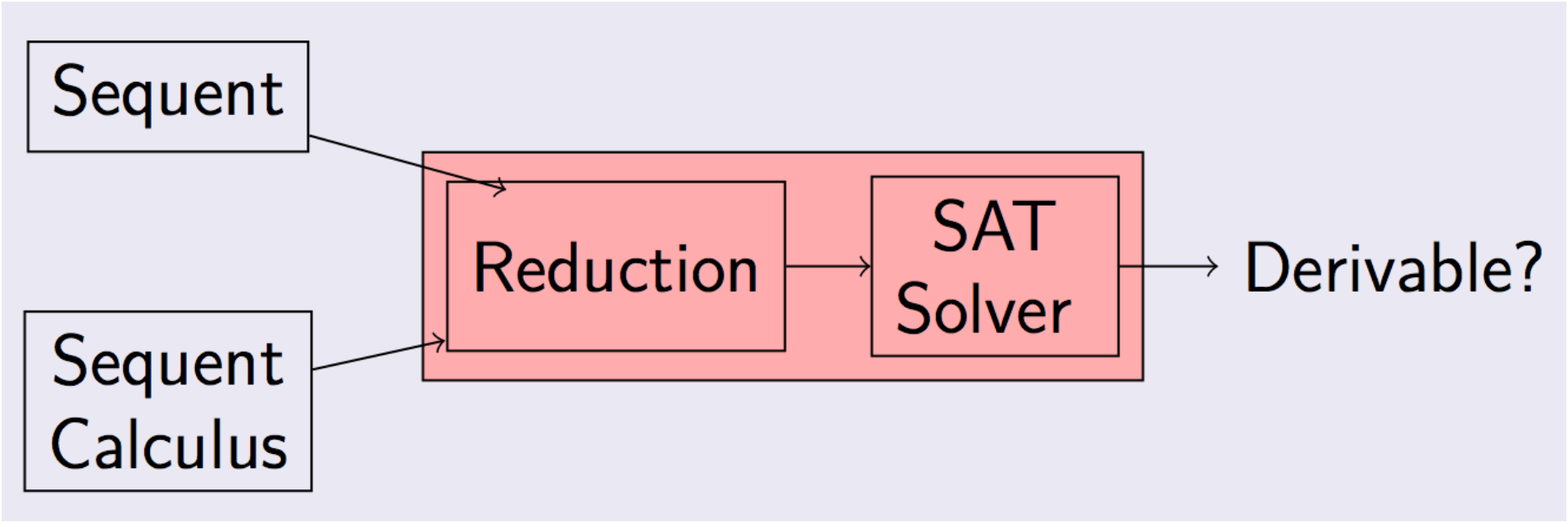
Uniform Decision Procedure



Uniform Decision Procedure

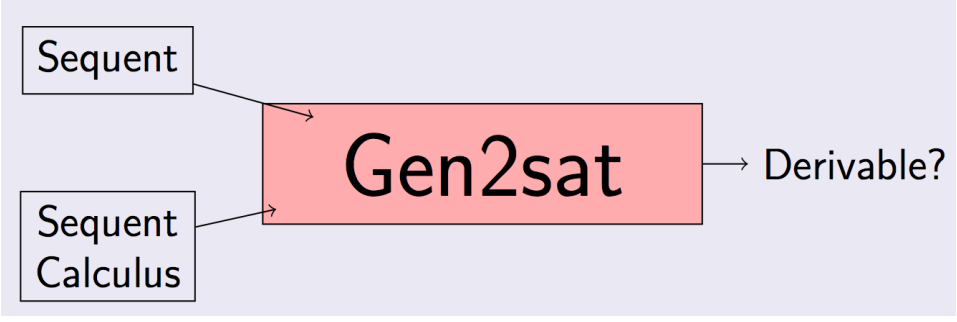


Uniform Decision Procedure



- Works for all pure $\circledcirc$-k-analytic sequent calculi
- Performs well
- Implementation relies on SAT-solver heuristics

Extended Decision Procedure



- $\mathbf{G}_{\{F,\mathrm{D}_F\}}$ admits the same reduction to SAT as $\mathbf{G}$
- Correctness is more challenging
- Implemented in Gen2sat as well

Tools

Main Tool: Non-deterministic Semantics

Semantic Frameworks

- Two-valued valuations
- Three-valued valuations
- Two-valued Kripke-models
- Three-valued Kripke models
- Non-deterministic matrices

General Recipe

- Formalize a syntactic hypothesis
- Prove a general soundness and completeness theorem
- Find a semantic hypothesis that entails the syntactic one
- Prove the semantic hypothesis
- Conclude the syntactic hypothesis

Semantics of Pure Calculi [Béziau 2001]

Bivaluations

A bivaluation is a function $v : \mathcal{WFF} \to \{-1, 1\}$

Satisfaction of sequents

$v \models \Gamma \Rightarrow \Delta$ if either $\exists A \in \Gamma . v(A) = -1$ or $\exists B \in \Delta . v(B) = 1$

G-legal bivaluations

A valuation is **G**-legal if it respects the "semantic reading" of **G**.

Example (Classical conjunction)

$$\frac{\Rightarrow A \quad \Rightarrow B}{\Rightarrow A \wedge B} \qquad \frac{A \Rightarrow}{A \wedge B \Rightarrow} \qquad \frac{B \Rightarrow}{A \wedge B \Rightarrow}$$

1. If $v(A) = 1$ and $v(B) = 1$ then $v(A \wedge B) = 1$
2. If $v(A) = -1$ then $v(A \wedge B) = -1$
3. If $v(B) = -1$ then $v(A \wedge B) = -1$

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G-legal bivaluations

A valuation is **G**-legal if it respects the "semantic reading" of **G**.

Example (Non-classical negations)

$$\frac{A \Rightarrow}{\Rightarrow \neg A} \qquad \frac{A \Rightarrow}{\neg\neg A \Rightarrow} \qquad \frac{\neg A \Rightarrow \quad \neg B \Rightarrow}{\neg(A \wedge B) \Rightarrow}$$

1. If $v(A) = -1$ then $v(\neg A) = 1$
2. If $v(A) = -1$ then $v(\neg\neg A) = -1$
3. If $v(\neg A) = -1$ and $v(\neg B) = -1$ then $v(\neg(A \wedge B)) = -1$

This semantics is non-deterministic.

Semantics of Pure Calculi [Béziau 2001]

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G-legal bivaluations

A valuation is **G**-legal if it respects the "semantic reading" of **G**.

Soundness and Completeness [Béziau '01]

s is provable in **G**

$$\Longleftrightarrow$$

s is satisfied by every **G**-legal bivaluation

Semantics of Pure Calculi [Béziau 2001]

Bivaluations

A bivaluation is a function $v : \mathcal{WFF} \to \{-1, 1\}$

Satisfaction of sequents

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G-legal bivaluations

A valuation is **G**-legal if it respects the "semantic reading" of **G**.

Soundness and Completeness

s is provable in **G** using $\mathcal{F} \subseteq WFF$

$\Longleftrightarrow$

s is satisfied by every **G**-legal bivaluation with domain $\mathcal{F}$

Proof of Analyticity Criterion

Syntactic Hypothesis	If **G** is coherent and directed then it is analytic
Soundness & Completeness	Bivaluations and partial bivaluations
Semantic Hypothesis	Every **G**-legal partial bivaluation with domain $sub(s)$ can be extended to a full one
Prove It!	Construct a general extension method for partial bivaluations
Conclude Original	☺☺☺

Semantics of Pure Calculi without cut [Lahav, Avron 2011]

Main Idea

The cut rule should not be sound.

$$\frac{\Rightarrow A \quad A \Rightarrow}{\Rightarrow}$$

Bivaluations [Béziau 2001]

A bivaluation is a function $v : \mathcal{WFF} \to \{-1, \quad 1\}$

Satisfaction of sequents

$v \models \Gamma \Rightarrow \Delta$ if either $\exists A \in \Gamma . v(A) \quad = \quad -1$ or $\exists B \in \Delta . v(B) = 1$

Soundness and Completeness

s is provable in **G** without cut

$\Longleftrightarrow$

s is satisfied by every **G**-legal trivaluation

Semantics of Pure Calculi without cut [Lahav, Avron 2011]

Main Idea

The cut rule should not be sound.

$$\frac{\Rightarrow A \quad A \Rightarrow}{\Rightarrow}$$

Trivaluations [Schütte 1958]

A trivaluation is a function $v : \mathcal{WFF} \to \{-1, 0, 1\}$

Satisfaction of sequents

$v \models \Gamma \Rightarrow \Delta$ if either $\exists A \in \Gamma . v(A) \in \{-1, 0\}$ or $\exists B \in \Delta . v(B) \in \{0, 1\}$

Soundness and Completeness

s is provable in **G** without cut

$\Longleftrightarrow$

s is satisfied by every **G**-legal trivaluation

Cut-admissibility vs. Analyticity

Syntactic Hypothesis	If **G** is analytic and directed then it admit cut-admissibility
Soundness & Completeness	Bivaluations and trivaluations
Semantic Hypothesis	Every **G**-legal trialuation can be transformed into a **G**-legal bivaluation
Prove It!	Construct a general method to eliminate the third value
Conclude Original	☺☺☺

Modal Analyticity

Syntactic Hypothesis	If **G** is analytic then so is $\mathbf{G}_X$
Soundness & Completeness	Kripke models
Semantic Hypothesis	If every partial bivaluation can be extended, then every partial Kripke model can be extended
Prove It!	Extend each possible world separately
Conclude Original	☺☺☺

Matrices

Logical Matrices

A matrix **M** consists of:

- a set $\mathcal{V}$ of *truth values*
- a subset $\mathcal{D}$ of *designated* truth values
- for each connective $\diamond$ of arity n, a "truth table" $\mathcal{O}(\diamond) : \mathcal{V}^n \to \mathcal{V}$

M-valuations

$$v(\diamond(A_1, \ldots, A_n)) = \mathcal{O}(\diamond)(v(A_1), , \ldots, v(A_n))$$

$$\begin{aligned} v(p) &= a_1 \\ v(q) &= a_2 \\ v(p \wedge q) &= a_2 \\ v(q \wedge p) &= a_3 \\ &\vdots \end{aligned}$$

When Does $T \vdash_{\mathbf{M}} A$?

For every **M**-valuation v:

$$v(A) \in \mathcal{D} \text{ for every } \psi \in \mathcal{T} \Longrightarrow v(A) \in \mathcal{D}$$

Non-deterministic Matrices

Non-deterministic Logical Matrices [Avron et al.'05]

A Nmatrix **M** consists of:

- a set $\mathcal{V}$ of *truth values*
- a subset $\mathcal{D}$ of *designated* truth values
- for each connective $\diamond$ of arity n, a "truth table" $\mathcal{O}(\diamond) : \mathcal{V}^n \to P^+(\mathcal{V})$

M-valuations

$$v(\diamond(A_1, \ldots, A_n)) \in \mathcal{O}(\diamond)(v(A_1), , \ldots, v(A_n))$$

$$\begin{aligned} v(p) &= a_1 \\ v(q) &= a_2 \\ v(p \wedge q) &= a_2 \\ v(q \wedge p) &= a_3 \\ &\vdots \end{aligned}$$

When Does $T \vdash_{\mathbf{M}} A$?

For every **M**-valuation v:

$$v(A) \in \mathcal{D} \text{ for every } \psi \in \mathcal{T} \Longrightarrow v(A) \in \mathcal{D}$$

Example: C_{min}

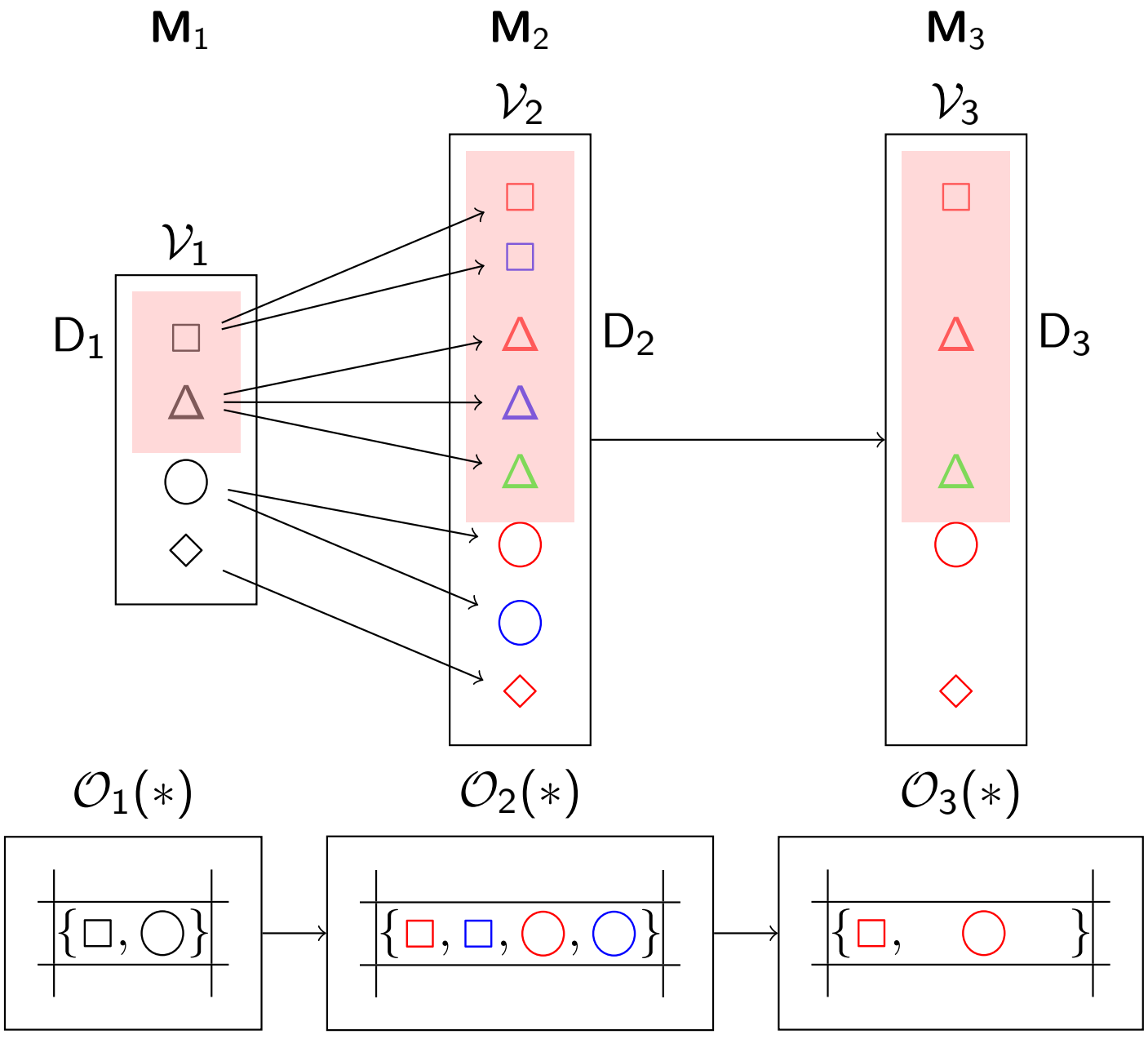
$\mathcal{O}(\wedge)$	t	$\top$	f
t	$\{t,\top\}$	$\{t,\top\}$	$\{f\}$
$\top$	$\{t,\top\}$	$\{t,\top\}$	$\{f\}$
f	$\{f\}$	$\{f\}$	$\{f\}$

$\mathcal{O}(\vee)$	t	$\top$	f
t	$\{t,\top\}$	$\{t,\top\}$	$\{t,\top\}$
$\top$	$\{t,\top\}$	$\{t,\top\}$	$\{t,\top\}$
f	$\{t,\top\}$	$\{t,\top\}$	$\{f\}$

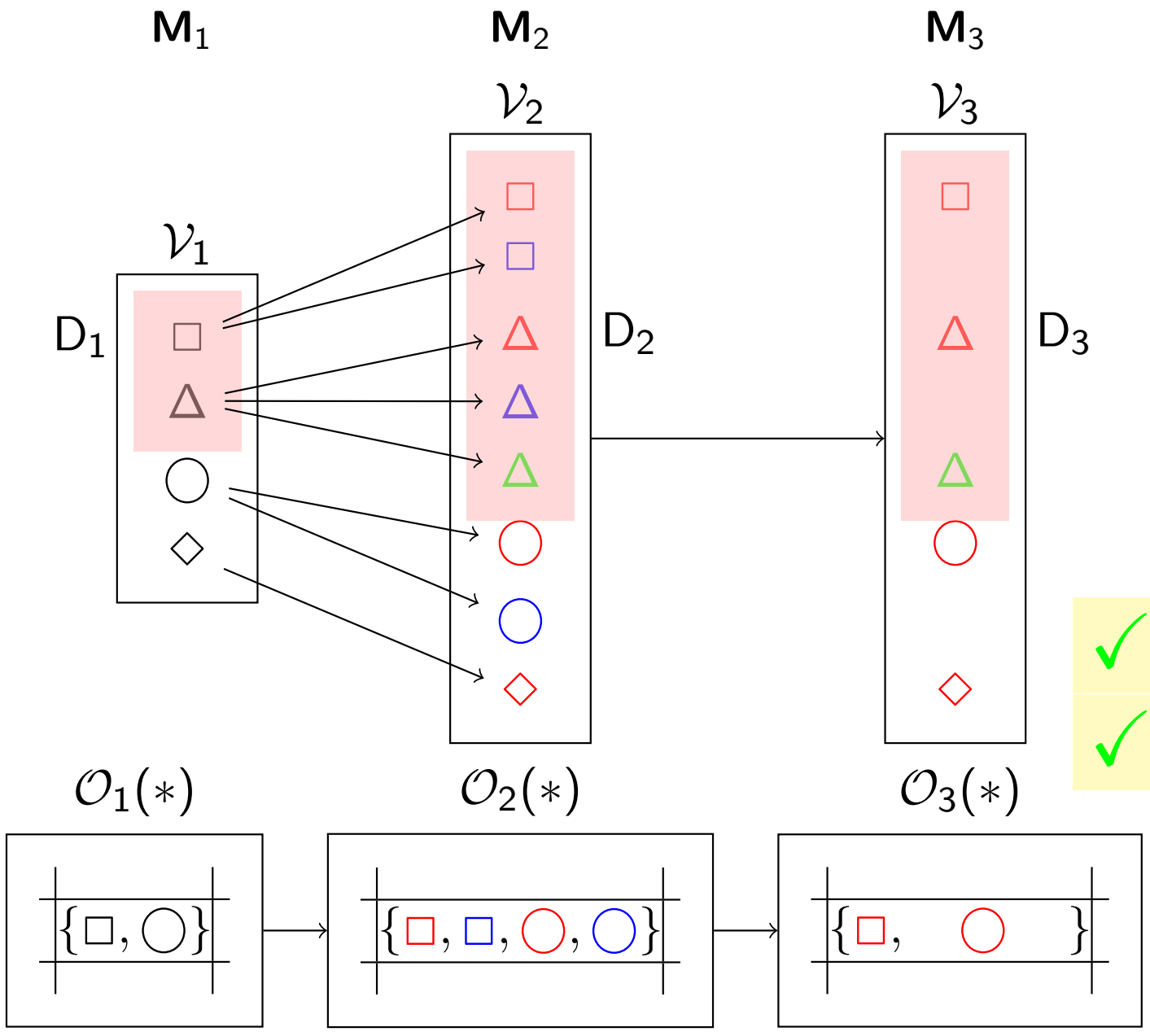
$\mathcal{O}(\supset)$	t	$\top$	f
t	$\{t,\top\}$	$\{t,\top\}$	$\{f\}$
$\top$	$\{t,\top\}$	$\{t,\top\}$	$\{f\}$
f	$\{t,\top\}$	$\{t,\top\}$	$\{t,\top\}$

	$\mathcal{O}(\neg)$
t	$\{f\}$
$\top$	$\{t,\top\}$
f	$\{t\}$

Rexpansions: Refined Expansions



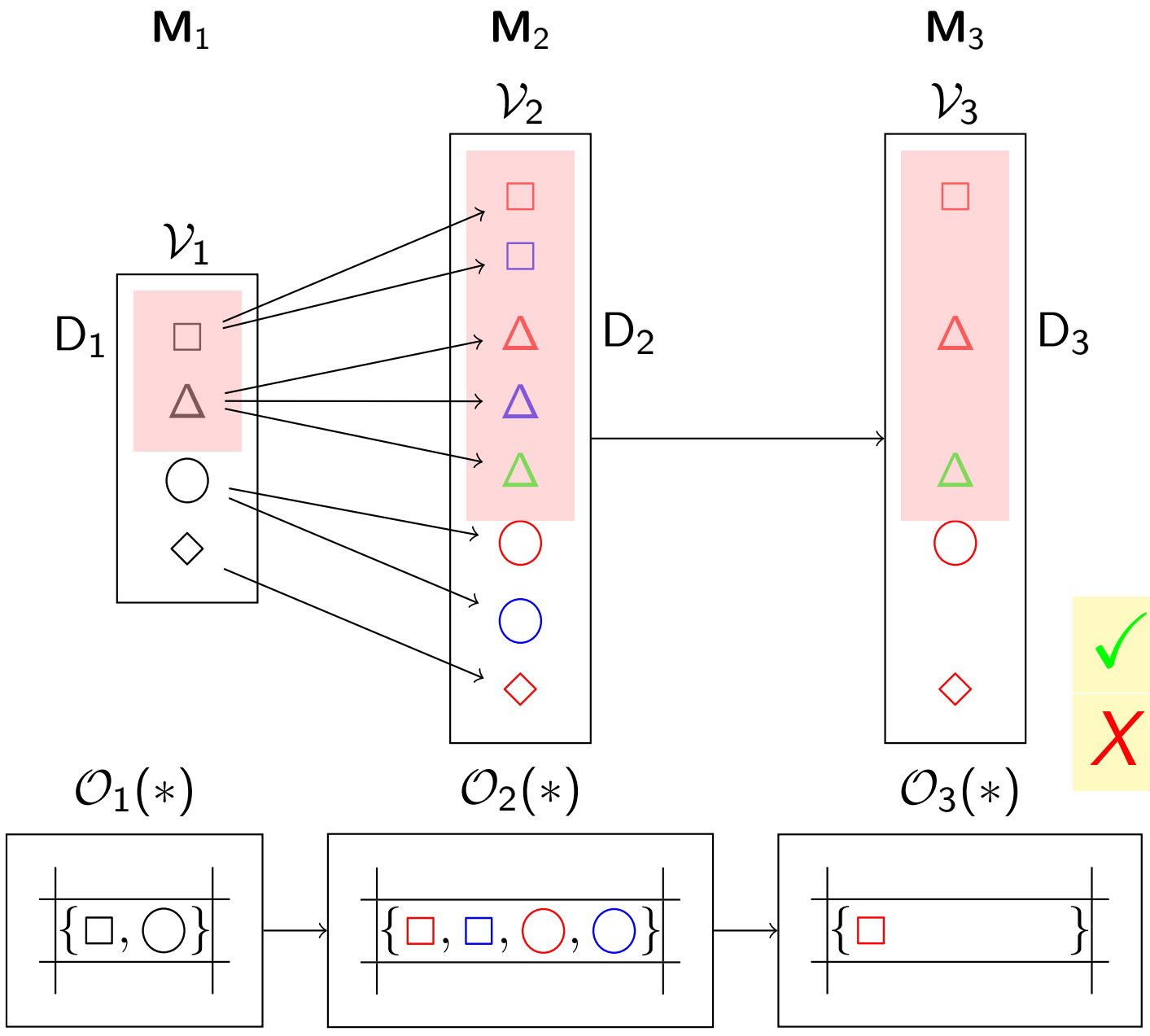
Rexpansions: Refined Expansions



✓ preserving

✓ strongly preserving

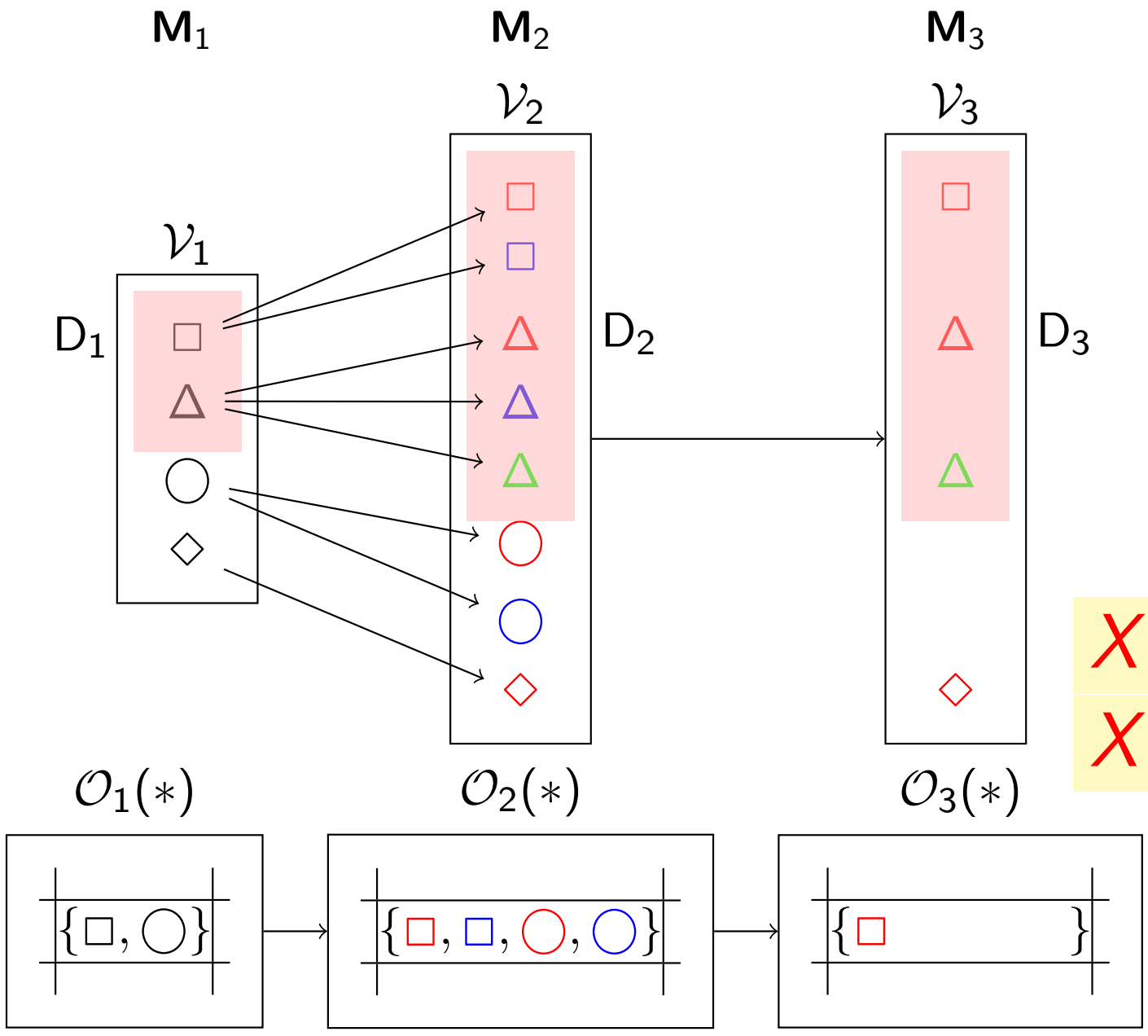
Rexpansions: Refined Expansions



✓ preserving

✗ strongly preserving

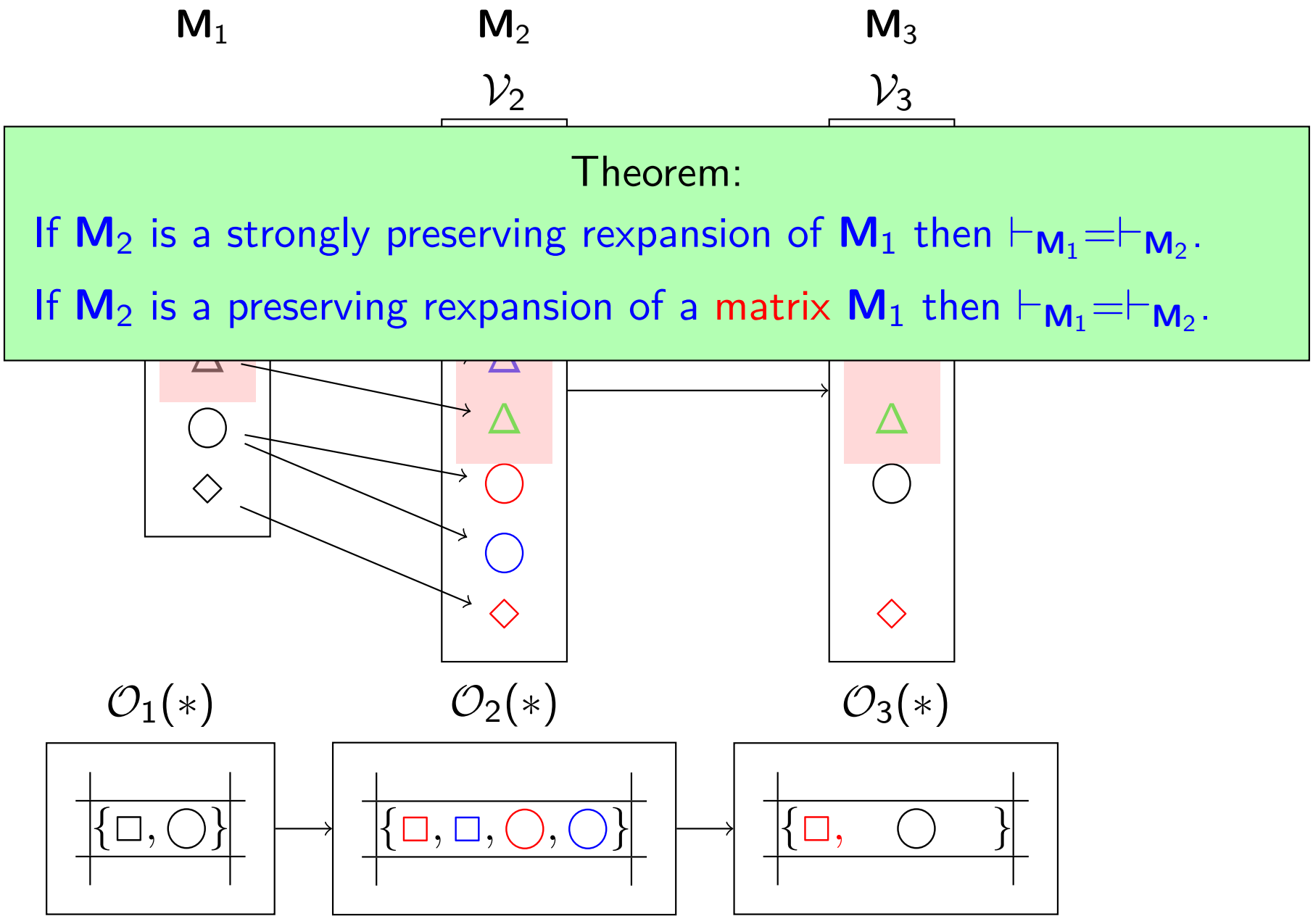
Rexpansions: Refined Expansions



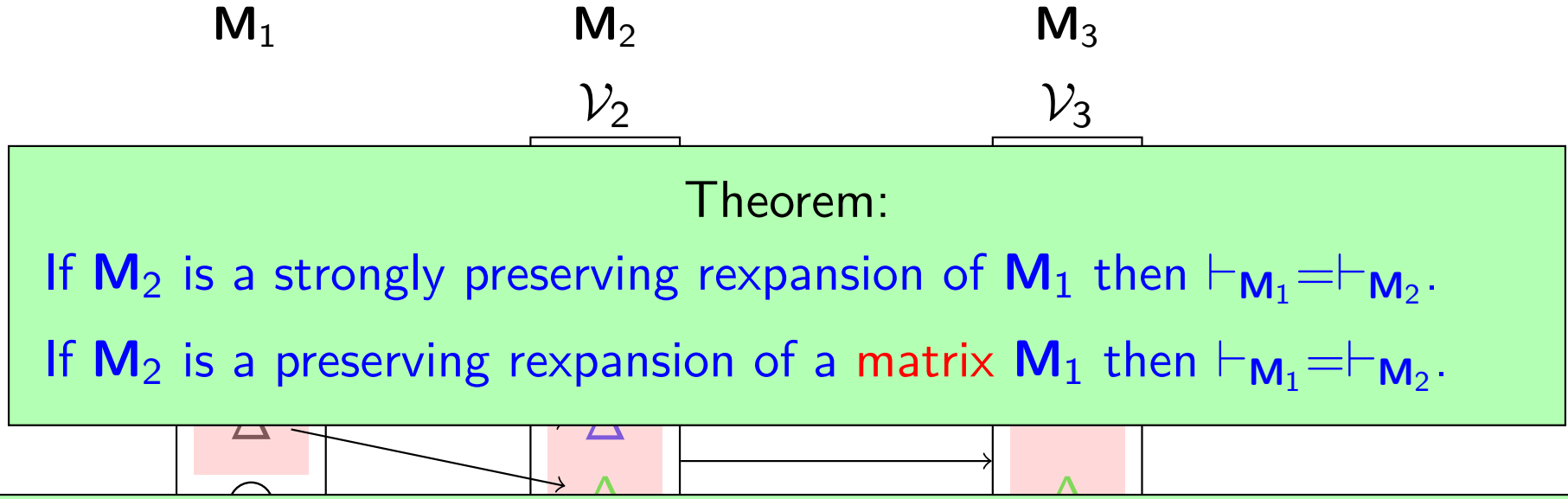
✗ preserving

✗ strongly preserving

Rexpansions: Refined Expansions



Rexpansions: Refined Expansions



Theorem:

If $\mathbf{M}_2$ is a strongly preserving rexpansion of $\mathbf{M}_1$ then $\vdash_{\mathbf{M}_1} = \vdash_{\mathbf{M}_2}$.

If $\mathbf{M}_2$ is a preserving rexpansion of a matrix $\mathbf{M}_1$ then $\vdash_{\mathbf{M}_1} = \vdash_{\mathbf{M}_2}$.

- Provides a method to obtain conservative extensions:
  - Start with an Nmatrix
  - Perform a rexpansion
  - Add a new connective
- Examples:
  - Paraconsistent conservative extensions of Gödel fuzzy logic
  - Conservative extensions of positive classical logic
- A “secret ingredient" in the construction of analytic proof systems

Conclusion

We have seen:

- A general investigation of families of calculi
  - Pure calculi
  - Pure calculi + modal operators
  - Intuitionistic calculi
- Results
  - Analyticity conditions
  - Decision procedure
  - Analyticity vs. Cut-admissibility
  - Construction Methods
- The proofs are semantic
- Rexpansions of Nmatrices

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Thank You!